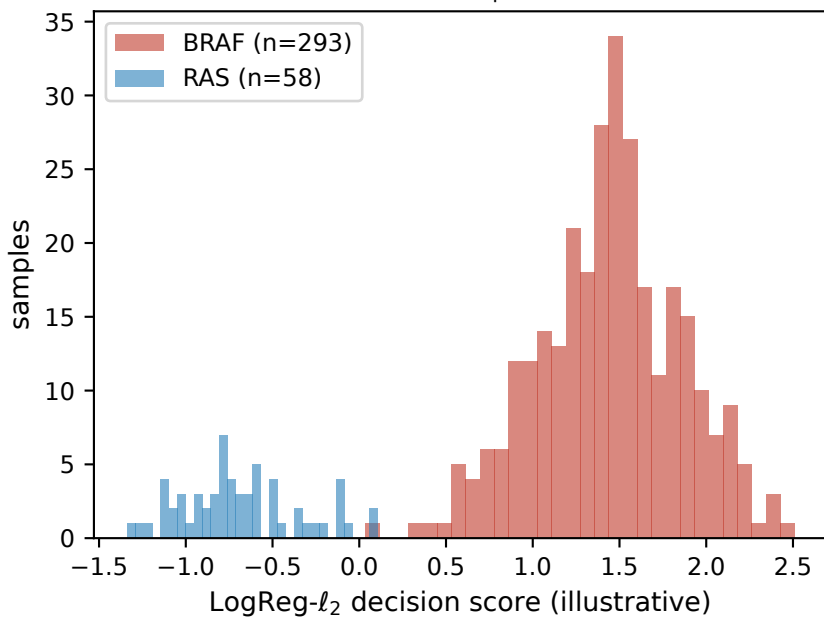
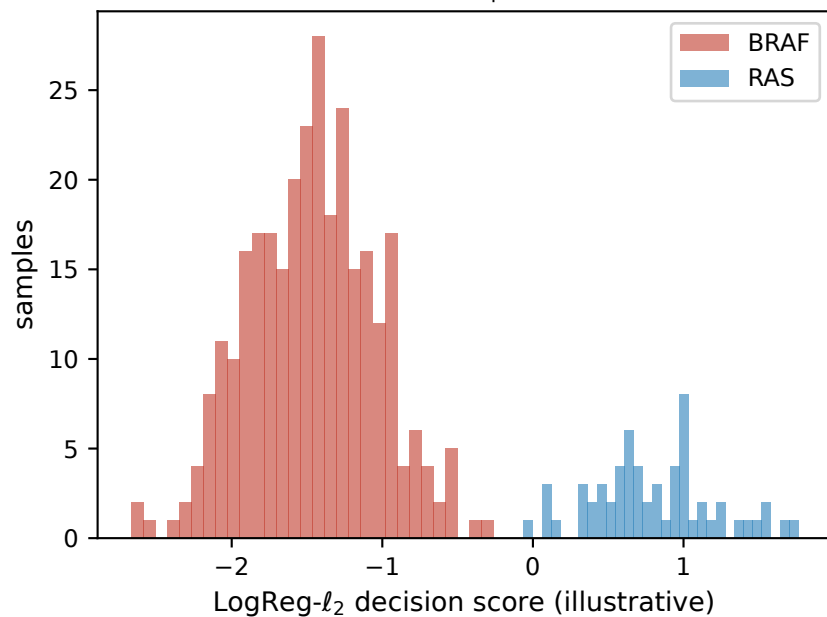


Geometry of the THCA label flip (panels A, B, C, D are schematic illustrations)

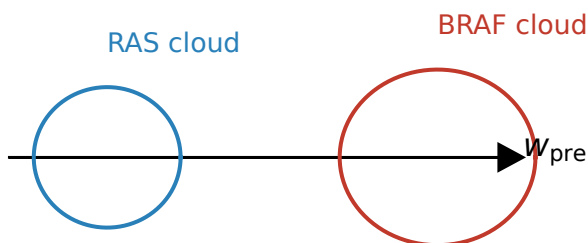
(A) Pre-ComBat score (schematic)
observed $AUC_{pre}=0.994$



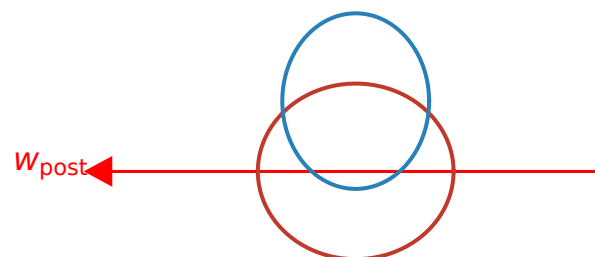
(B) Post-ComBat score (schematic)
observed $AUC_{post}=0.006$



(C) Pre-ComBat (schematic): discriminant aligned with cohort axis $\mathcal{V}_B$



(D) Post-ComBat (schematic): residual-batch eigenvector with flipped sign



classes collapsed after ComBat shrinkage